

Comprehensive Evaluation of a Bimanual Tangible Simulator for Microsurgical Clipping

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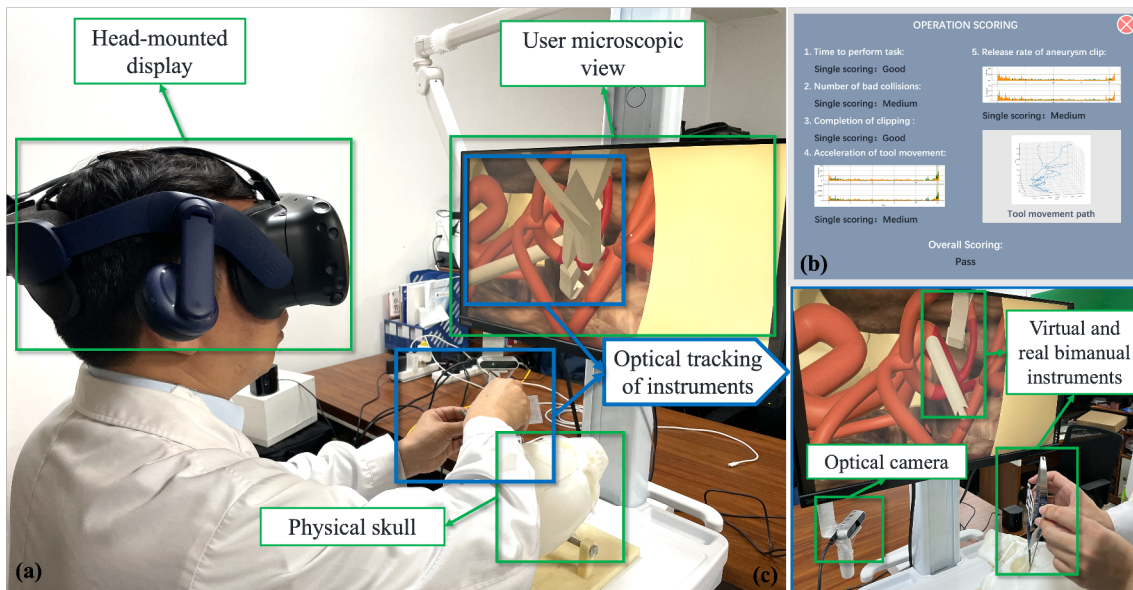


Figure 1: Our microsurgical clipping simulator. (a) A bimanual tangible interface for virtual microsurgical clipping of intracranial aneurysms. (b) Quantitative assessment of the operative skills.

ABSTRACT

Virtual rehearsal systems have emerged as a crucial tool in microsurgery, yet current approaches predominantly emphasize the evaluation of system validity while overlooking the operator's skill assessment. To tackle the above issue, this study first develops a space-aware bimanual tangible interface for high-fidelity microsurgical clipping simulation of intracranial aneurysms. After that, we evaluate the validity of this interface through user studies with six neurosurgeons and seven residents. Building upon this interface, we design an objective and quantitative assessment scale. Through consultations with neurosurgeons, we employ 15 construct validity metrics to analyze the operation accuracy and kinematic dexterity. These metrics are systematically consolidated into five standardized scoring criteria to evaluate users' skills. The results demonstrate the effectiveness of our prototype and exhibit promising potential to be applied in real neurosurgical rehearsal.

Index Terms: Virtual rehearsal, bimanual tangible interface, skill

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1 INTRODUCTION

Intracranial aneurysms (IAs) are associated with high morbidity and mortality upon rupture, and microsurgical clipping (MC) remains a critical treatment option for IAs [16]. During an MC procedure, the neurosurgeon first dissects the surrounding brain tissue to expose the aneurysm, then selects an appropriate aneurysm clip based on the observed morphology and places it to occlude the aneurysm neck. However, due to factors such as the complexity of intracranial anatomy, the limited microscopic field of view, and confined operating space, MC carries extremely high risks. This places great demands on neurosurgeons' spatial awareness and intraoperative decision-making abilities. Conventional surgical training has inherent limitations, such as high training costs, a lack of operational feedback, and difficulty in replicating risky scenarios. Consequently, virtual surgical training and rehearsal systems, through real-time interaction and repeated operations with digital twins of patients' anatomy, have gradually become a core method for improving the cost-effectiveness of surgical skill training [4, 9, 7].

[16] and [7] conducted a systematic review of aneurysm clipping simulators, including several studies that leveraged established commercial platforms for clinical trials. Several studies have systematically validated the efficacy of the Surgical Theater platform in neurosurgical training and preoperative planning [11, 14, 4].

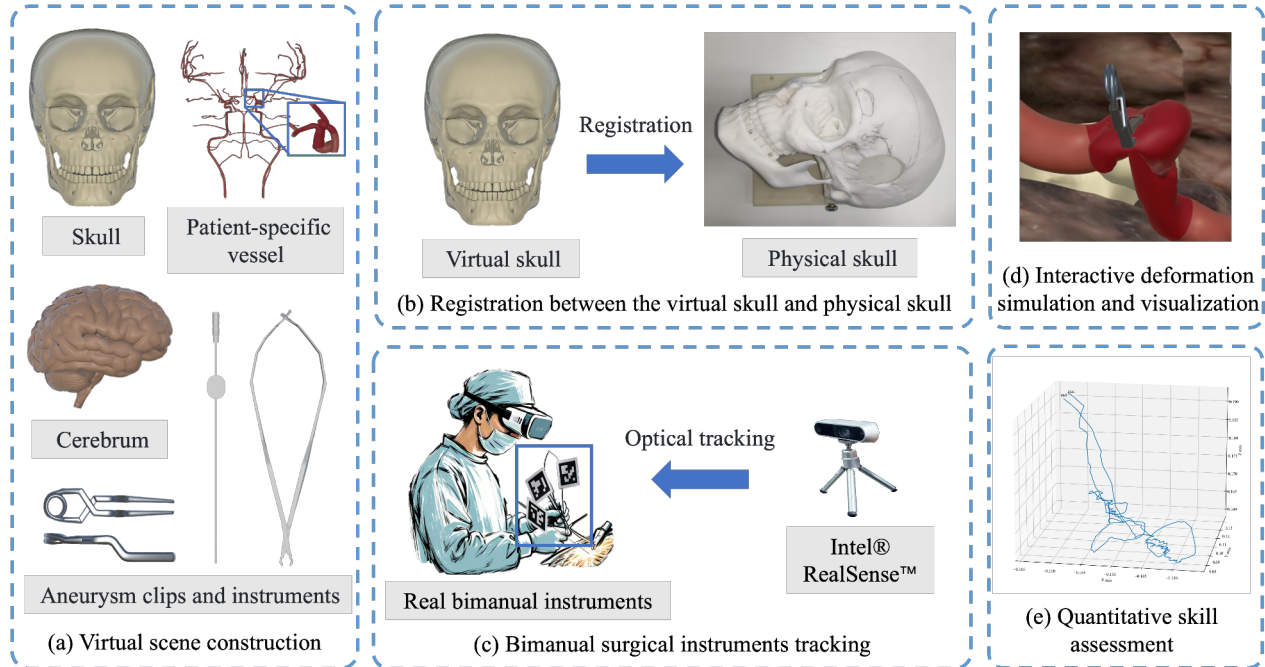


Figure 2: The pipeline of our MC simulator. (a) Construct the geometry models for the virtual surgical scenario. (b) Register the virtual skull to the physical one. (c) Track the bimanual surgical instruments with a depth camera. (d) Perform the clipping operation and visualize the scene. (e) Assess the operative skills with quantitative metrics.

These collective findings robustly demonstrate VR's multidimensional benefits in neurosurgical training and clinical practice. [8] and [13] employed the Dextroscope platform for 3D reconstruction and surgical simulation, confirming its utility in identifying complex vascular variations and optimizing surgical strategies. However, both studies acknowledged limitations, including insufficient clipping simulation fidelity and the absence of haptic feedback.

Recent studies have explored the development of virtual aneurysm clipping simulators using open-source platforms. Marinho et al. [10] leveraged Blender to create a virtual surgical environment, validating its efficacy for IA clipping visualization through high-resolution 3D image processing. Shono et al. [12] utilized Unity to develop a VR clipping simulator incorporating deformable brain tissue and aneurysm vessel modeling. Allgaier et al. [2] implemented a VR training system on Unity and employed VR Ink as the input device. However, the absence of a 3D-printed skull model for spatial reference exacerbated the device's controllability issues. Teodoro et al. [15] developed a hybrid aneurysm clipping simulator using the SOFA framework to model tissue biomechanics, though its virtual parenchymal retraction exceeded realistic behavior. Similarly, Civilla et al. [5] created an AR simulator with the SOFA framework and designed a finite element modeling (FEM) method for deformation simulation. T Gmeiner et al. [6] implemented a VR clipping simulator integrating real surgical tools with haptic feedback, yet restricted instrument mobility reduced the operative immersion. Zhang et al. [17] focused on preoperative planning via a VR-guided Focused Sylvian Approach, enhancing spatial assessment of aneurysm morphology and adjacent structures during the preoperative planning stage.

However, most existing studies primarily evaluate simulator performance through user surveys, lacking objective and quantitative metrics to assess the skill of users. To address this limitation, we first present a high-fidelity MC simulator based on a bimanual tangible interface, as shown in Fig. 1 (a). And then for evaluation

and validation, we consider two aspects. On one hand, we invite six neurosurgeons and seven residents to test our system. Then we ask them to rate the system's performance by subjective measures. On the other hand, based on validity assessment criteria [1], we record participants' operation kinematics and time, among other parameters, and analyze the stability of their operations and inter-individual differences through kinematic dexterity, enabling a quantitative evaluation of surgeons' performance.

2 METHODS

2.1 Design of the MC simulator

Our MC simulator aims to provide the procedure simulation of clipping a patient-specific aneurysm, including the interactive deformation between the heterogeneous brain tissues and bimanual surgical instruments. Fig. 2 shows the pipeline of our MC simulator. First of all, to generate the virtual surgical scenario, we reconstruct standardized skull and cerebrum models, patient-specific brain vascular models, as well as surgical instrument models, including clip holder, aspirator, and aneurysm clips, as illustrated in Fig. 2 (a). Then, we use a 3D printed physical skull to enhance users' spatial perception during the operation simulation, as shown in Fig. 2 (b). The physical skull is fixed using a bracket with screws. For each patient-specific case, we first adjust the position of the physical skull and then register the virtual skull with it. For real-time human-computer interaction, we develop a bimanual tangible interface by using an optical camera to track the real microsurgical instruments, as shown in Fig. 2 (c). We first attach one ArUco marker to the aspirator and two ArUco markers to the clip holder, and then use OpenCV to detect and calculate the markers' pose. Fig. 1 (a) also shows our bimanual tangible interface. To improve the realism of the clipping procedure, we simulate the deformation of heterogeneous brain tissues when the surgical instruments touch them using the algorithm in [3]. We perform visual rendering of

the clipping procedure in the microscopic view, as shown in Fig. 2 (d). Finally, we integrate the above functions into a virtual surgery simulator and record user operation data for quantitative analysis, enabling objective assessment of users' surgical skill, as shown in Fig. 2 (e). In addition, we employ a head-mounted display (HMD) to simulate the microscopic view, which provides an immersive surgical operating experience.

2.2 Skill assessment scale design

We design a user-friendly interactive interface for the rehearsal system, which allows users to switch aneurysm cases, change view-points, and perform other functions using a controller. The simulator preloads three intracranial aneurysm cases and five available aneurysm clips for selection. During the microsurgical clipping procedure, the user can first open the microscopic view to observe the morphology of the aneurysm. Then, they can open the control panel and select an appropriate aneurysm clip using the "Switch Clip" function. Afterward, the control panel is closed, and the user can grasp the real surgical instruments with both hands to begin the clipping operation.

We record the kinematic data of instrument manipulations. Upon complete clip release and closure, the system automatically analyzes these kinematic parameters, which are presented in detail in Sec. 3.2. Through discussion with neurosurgeons, we established five evaluation metrics for surgical skill assessment, each graded on three scales as good, medium, and poor: (1) Operational proficiency is evaluated based on the time duration of the clipping procedure. (2) Spatial perception ability is determined by counting tool-tissue collision occurrences. (3) Completion of clipping is assessed by measuring the average distance between the aneurysm neck vertices and the aneurysm clip. (4) Instrument control stability is measured through movement acceleration analysis in confined surgical spaces. (5) Clipping stability is judged by the speed of clip releasing. Finally, the simulator displays a skill scoring panel, as shown in Fig. 1 (b).

3 EXPERIMENT AND ASSESSMENT

3.1 Environment description

The rehearsal system is deployed on a desktop computer with an Intel(R) Core(TM) i9-10900X CPU @ 3.70GHz (20 CPUs), a NVIDIA GeForce RTX 3090 GPU, and 128GB of RAM. The HMD is an HTC Vive Pro headset with a 110° field of view. The development engine is Unity3D with version 2021.3.21f1, which is supported by SteamVR 2.8.0 and its Unity Plugin 2.0.10. The Obi physics engine version 6.4.0 in Unity is used for the deformation simulation of brain tissue. The surgical instruments involve one clip holder, one aspirator, and five aneurysm clips (FT742T, FT746T, FT752T, FT758T, and FT761T), which are modeled as 3D meshes and imported into Unity3D. The Intel@RealSense™ depth camera D435i is equipped for the marker-based instruments tracking, which offers a maximum resolution of 1280x720 and a refresh rate of over 30Hz. OpenCV version 4.8.1 is utilized for marker detection and pose calculation.

3.2 User study

We invited six neurosurgeons and seven residents to test the rehearsal system and conducted a post-task questionnaire. All participants voluntarily enrolled in this study and signed informed consent. The local ethics committee at the hospital approved the study. The demographic data is presented in Tab. 1

All participants were required to wear an HMD and perform the intracranial aneurysm clipping for the three cases using bimanual instruments, followed by answering a questionnaire. We measured the perceived workload when performing the MC task in our simulator by using NASA TLX, as shown in Fig. 3. The bimanual interaction imposed a relatively high workload on the participants.

Table 1: Demographic data for the participants performing our rehearsal system.

	Chief surgeons	Attending surgeons	Surgical residents
Number of participants	3 (23%)	3 (23%)	7 (54%)
Mean age (yr) $\pm$ SD ¹	44.7 $\pm$ 5.0	36.7 $\pm$ 2.6	26.9 $\pm$ 1.8
Sex			
Male	3 (100%)	3 (100%)	4 (57%)
Female	0 (0%)	0 (0%)	3 (43%)
Level of training			
Neurosurgery resident			
PGY 1-3	-	-	6
PGY 4-6	-	-	1
Spine fellows			
Neurosurgery	3	3	-

¹SD = standard deviation.

Some reported that due to certain delays and inaccuracies in instrument tracking, when they focused on manipulating the clip holder to occlude the aneurysm, they often neglected control of the aspirator-holding hand. This resulted in frequent trembling of the aspirator instrument in the virtual microscopic field of view, thereby increasing the operational burden. Overall, there is broad agreement among all participants that the bimanual interaction interface we developed closely mimicked real surgical procedures.

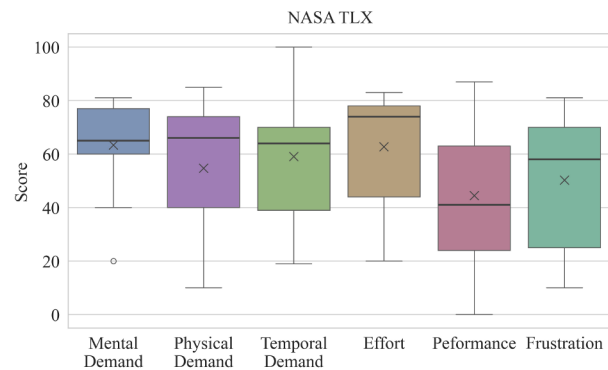


Figure 3: The measurement of workload for the MC task in our simulator.

3.3 Quantitative assessment

For the quantitative assessment, we applied a series of multivariate analyses of variance to the kinematic dexterity metrics over the three aneurysm cases and two groups (resident vs. neurosurgeon), called construct validity [1]. The simulator automatically recorded the data of the clipping operations from the two groups, and the construct validity analysis results are presented in Tab. 2, where **P** values in bold are statistically significant.

For the clipping task performance, we measured the time for completing the task (Time), the distances between the final positions of the clip with respect to the aneurysm neck (Clip-neck distance) and dome (Clip-dome distance), the aneurysm clip deployment angular velocity (C-Velocity) and acceleration (C-Acceleration), and the number of tool-tissue collisions (Collision counts). The results demonstrate statistically significant differences between the two groups in the metrics of Clip-neck distance, C-velocity, C-acceleration, and Collision counts. Clip-neck distance

Table 2: Comparing the performance of the aneurysm clipping task and its kinematic dexterity during task execution between the resident and neurosurgeon groups.

	Resident			Neurosurgeon			P
	LACom	LOptA	LPCom	LACom	LOptA	LPCom	
Task performance							
Time, s	28.88±3.20	24.45±1.46	25.88±5.14	29.10±5.74	23.83±2.82	28.15±4.46	NP
Clip-neck distance cm	0.06±0.01	0.08±0.01	0.06±0.01	0.05±0.01	0.07±0.01	0.03±0.01	<0.001
Clip-dome distance cm	0.08±0.03	0.10±0.02	0.06±0.02	0.09±0.01	0.10±0.02	0.03±0.01	NP
C-velocity, deg/s	2.16±0.44	1.45±0.35	1.92±0.98	1.17±0.24	0.84±0.12	1.13±0.17	<0.001
C-acceleration, deg/s ²	356.36±75.74	234.31±63.08	361.44±201.62	199.37±50.64	130.68±21.54	212.89±41.00	<0.001
Collision counts	3.86±1.07	5.00±1.15	3.71±1.70	2.17±1.17	2.33±1.03	2.00±0.89	<0.001
Kinematic performance							
Width X, cm	8.21±3.83	4.51±2.31	3.48±2.71	5.04±3.09	7.43±3.19	2.43±1.33	NP
Width Y, cm	5.60±2.78	3.53±1.16	4.39±2.84	4.11±1.68	4.63±1.50	2.88±2.86	NP
Width Z, cm	4.69±2.06	5.21±2.33	3.47±2.33	2.98±1.99	1.54±0.53	2.19±1.25	<0.001
Path length, cm	40.17±5.12	29.20±4.46	28.67±9.21	27.80±11.26	28.62±11.32	24.68±6.08	NP
Velocity, cm/s	1.40±0.19	1.19±0.14	1.09±0.12	0.97±0.42	1.22±0.53	0.87±0.16	NP
Acceleration, cm/s ²	201.45±28.73	203.29±26.16	184.36±12.36	121.84±38.10	114.83±42.48	148.59±21.01	<0.001
Jerk, cm/s ³	53843±9227	59380±8776	54918±3689	27292±8148	20450±7728	43421±5424	<0.001

can serve as an indicator of clipping completion. C-velocity and C-acceleration reflect the operational stability of clipping. Collision counts evaluate the spatial perception ability of the user.

For the kinematic dexterity performance, we measured the path length of the clipping operation and its corresponding spatial coordinate range as Width X, Width Y, and Width Z. Based on the measured values, we calculated the velocity, acceleration, and jerk of the tool movements. The results in Tab. 2 demonstrated no significant differences between the two groups in metrics such as the width along the X-axis and Y-axis, path length, and velocity, attributable to the enhanced spatial perception by using a physical skull. However, because of extensive experience in judging the depth of aneurysms and caution in instrument insertion, the width of neurosurgeons' operation space along the Z-axis is significantly shorter than residents. Additionally, neurosurgeons significantly outperformed residents in instrument movement acceleration and jerk metrics. This indicates experienced neurosurgeons exhibit superior stability and skill level during microsurgical clipping procedures.

Through the above analysis and consultations with neurosurgeons, we ultimately selected five key metrics for surgical skill assessment: (1) Time to perform task (Time). (2) Number of bad collisions (Contact num.). (3) Completion of clipping (Clip-neck distance). (4) Acceleration of tool movement (Acceleration and Jerk). (5) Release rate of aneurysm clip (C-velocity and C-acceleration). As illustrated in Fig. 1 (b), the simulator automatically evaluates participant performance based on these five metrics upon completion of the clipping simulation.

4 CONCLUSION

This paper presents a bimanual tangible interface for microsurgical clipping of intracranial aneurysms, incorporating an objective scoring mechanism to assess users' surgical skills. User studies demonstrate the interface's potential to be applied in neurosurgical operation rehearsal. In particular, the assessment metrics derived from quantitative construct validity analysis provide more objective and intuitive feedback on skill performance in medical training.

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